Great Learning

5-6 minutes

Business Applications with Agentic AI

Problem Statement: Financial Compliance Agent for Trade Surveillance

Business Context

A mid-sized financial institution must monitor insider trading to comply with SEC regulations and internal policies. Corporate insiders (executives, directors, and 10%+ shareholders) are required to report trades through SEC Forms 3, 4, and 5 within two business days. Compliance teams currently review thousands of insider trades manually each quarter. This requires checking SEC EDGAR filings, internal authorization limits, and historical trading records, while also maintaining clear audit trails for regulators.

Traditional rule-based surveillance systems rely on basic thresholds and generate 60-70% false positives because they lack context. They cannot analyze an insider's full trading history, detect complex timing patterns (such as trades before earnings announcements), or clearly explain why a transaction was flagged. As a result, compliance teams spend most of their time investigating false alerts and may miss more sophisticated violations.

A simple rule-based or classification system is insufficient because regulatory compliance requires transparent, explainable, multi-step reasoning backed by data from multiple sources. The institution needs an AI-powered compliance agent that can:

- automatically collect and combine data from SEC EDGAR and internal systems
- apply role-based trading limits and compliance rules with clear reasoning
- detect complex patterns across multiple transactions and time windows
- validate findings across multiple sources to reduce false positives
- produce clear, audit-ready investigation reports with full evidence trails

Objective

The objective is to build an agentic compliance investigation system that automates the end-to-end analysis of transactional disclosures from SEC EDGAR data to accurately differentiate legitimate regulatory violations from false positives. The system applies systematic compliance evaluation using role-based authorization frameworks, historical pattern analysis, quarterly volume tracking, and timing anomaly detection. It generates audit-ready investigation reports with complete evidence chains, grounded in verifiable data and clear, regulator-friendly explanations.